

Refactoring main.py

Contributions:

Khalid Albabtain – No Contributions

Tobey Alcock - No Contributions

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Jonathan Badger – Refactoring, Testing, Mini-report write-up

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Refactoring:

- 'flat_tax' variable removed, as it is not used in main.py
- 'generate_report()' has bad variable names:
 - X refactored to: 'purchases'
 - Y refactored to: 'customer_name'
 - Z refactored to: 'item'
 - A refactored to: 'total'
- 'generate_report()' clean-up if, else logic. As it is quite long and can be simplified using a switch statement
- 'generate_report()', move discount printing into a separate function called 'print_discount'
- Remove 'calculate_shipping_fee_for_heavy_items()' as it is not used

Additional Testing:

- Made sure all functions are tested, added tests for:
 - add_purchases()
 - print_discount()
- We also increased test coverage by adding tests for:
 - The report of an item with a prices lower than the tax threshold

As part of the refactoring, it was noted that the 'shipping fee' calculation could be calculated in one function. We already removed the shipping fee calculation for heavy items as that is already done in the 'calculate_shipping_fee' method. However, it was not something we felt we could refactor effectively as we were uncertain of how the shipping fee for fragile and weighted items would interact.